Steve Carlos Montenegro Aguilar

07453301407 | steveaguilarM@gmail.com | linkedin.com/in/sm-aguilar | github.com/CodedBySteve

Education

Birmingham City University — Birmingham, UK
BSc Honours in Computer Science | First Class — Sept 2022 – July 2026

Cadbury Sixth Form College — Birmingham, UK

- BTEC Level 3 National Extended Diploma in IT | Distinction
- BTEC Level 3 National Extended Certificate in Engineering | Merit
- Extra Unit - Further Engineering Mathematics | Distinction
- A level Physics | Pass
- AS level Mathematical Studies | A — Sept 2019 – July 2021

Experience

Kitchen Porter — Nov 2021 – Aug 2022
The Alchemist — Birmingham, UK

- Maintained kitchen cleanliness and supported staff in a fast-paced work environment
- Followed hygiene and safety standards while managing time-sensitive tasks
- Gained experience in teamwork, discipline, and workplace responsibility

Projects

Text Based adventure game | Python, JSON, Git, Declarative programming

- Developed a command-line adventure game in Python featuring branching storylines, player choices, combat sequences, inventory management and multiple endings.
- Designed a state management system using Python dictionaries and flags to track player progression, locations, inventory, events and game outcomes.
- Implemented an interactive exploration and grid based maze with navigation, traps and checkpoints.
- Developed JSON save/load functionality to persist player state, inventory, explored locations and unlocked game areas between sessions.

Just Travelin' | HTML, CSS, PHP, SQL, Xampp, JavaScript

- Developed a responsive travelled theme website as part of a team using HTML, CSS, SQL, JavaScript.
- Designed the SQL database, established data retrieval and importing within the website and database using PHP, for managed dynamic content.
- Delivered project documentation and usability report.

VirtuFlex – Robotic arm & smart glove | C++, Arduino, electronics

- Lead member of my team and programmer of the electronics enabling mobility capabilities of a robotic arm as well as the response behaviours based on a smart glove, using Arduino IDE to program in C++; I established rules for how motors (servo motors and stepper motors) respond based on the sensor readings (flex sensors and accelerometer).
- Took part in the design and further details of the project considering UN sustainability goals and market interests.
- Coordinated and communicated with my team to cover project goals and support progression, avoiding conflicting clashes of task dependencies of another that is assigned to every member.

Technical Skills

Languages: Java, Python, C++, SQL, HTML/CSS, PHP, JavaScript
Frameworks: JUnit, FastAPI, React
Developer Tools: Git, VS Code, PyCharm, IntelliJ, Eclipse
Libraries: pandas, NumPy, Matplotlib, Pygame, Sklearn